Embedded Systems Course – Introduction to Analog to Digital Converters (ADC)

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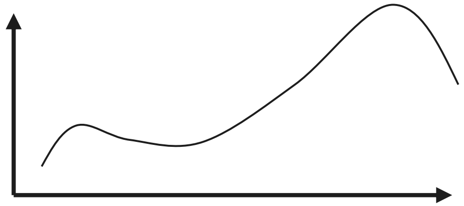
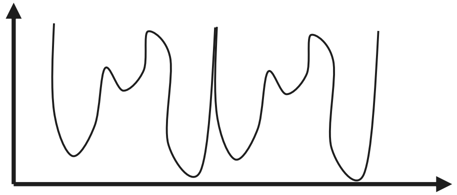
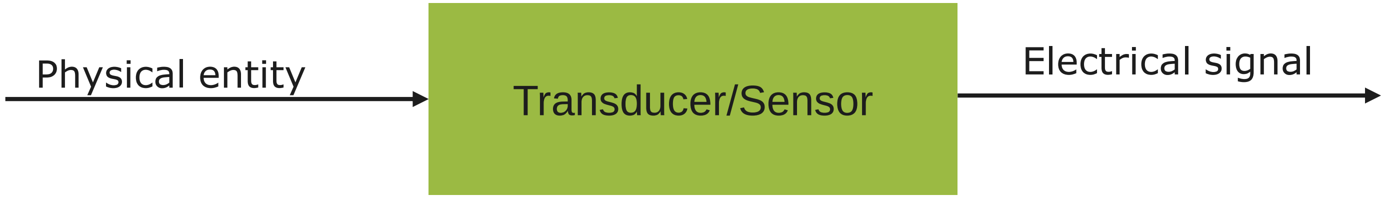
What is a signal?

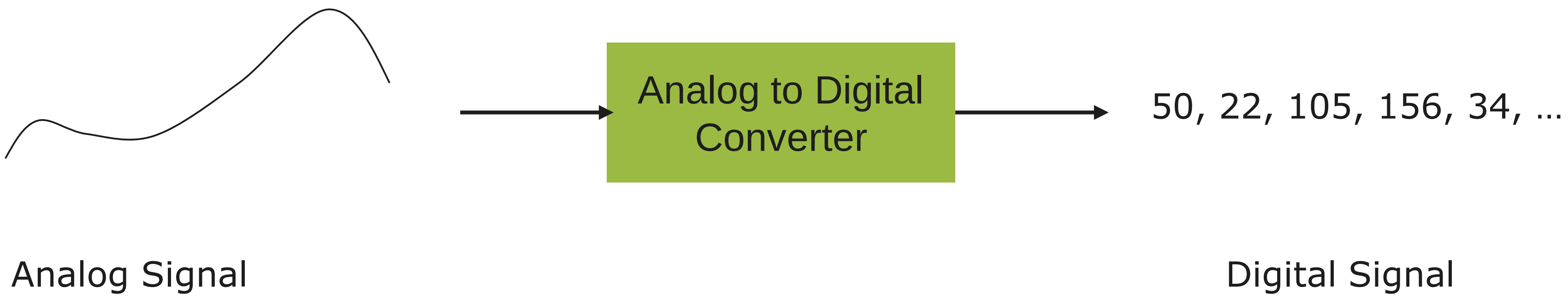
A physical quantity of interest which is a function of time







Digital representation of a signal



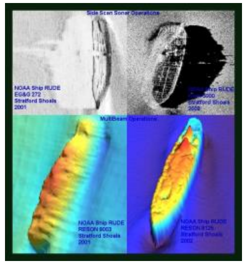
Practical use of Digital signal processing

Use	Techniques
Filtering	FIR, IIR, Convolution
Spectral Analysis	Windowing, DFT, FFT
Frequency conversion	Interpolation, Decimation
Correlation	Auto-correlation, Cross-correlation

Some applications of DSP

Communication

Optometry



Space and Sea exploration

Bio-Medical engineering

Radar

Describing signals in time domain

Notation

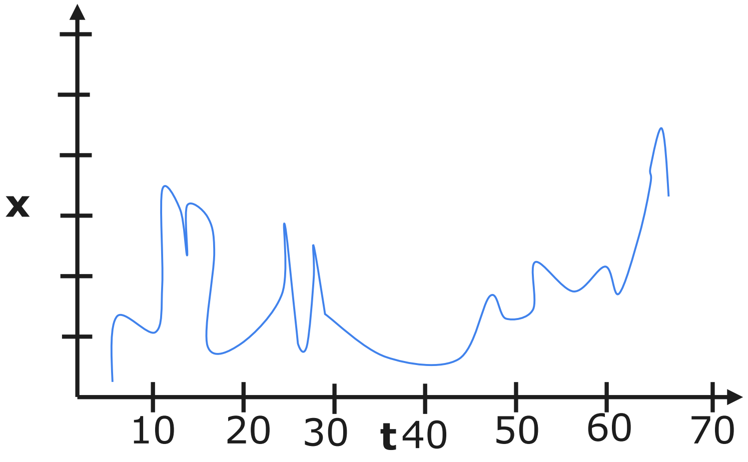
s(t), a(t), x(t), p(t) etc.

Function – Always in lower case

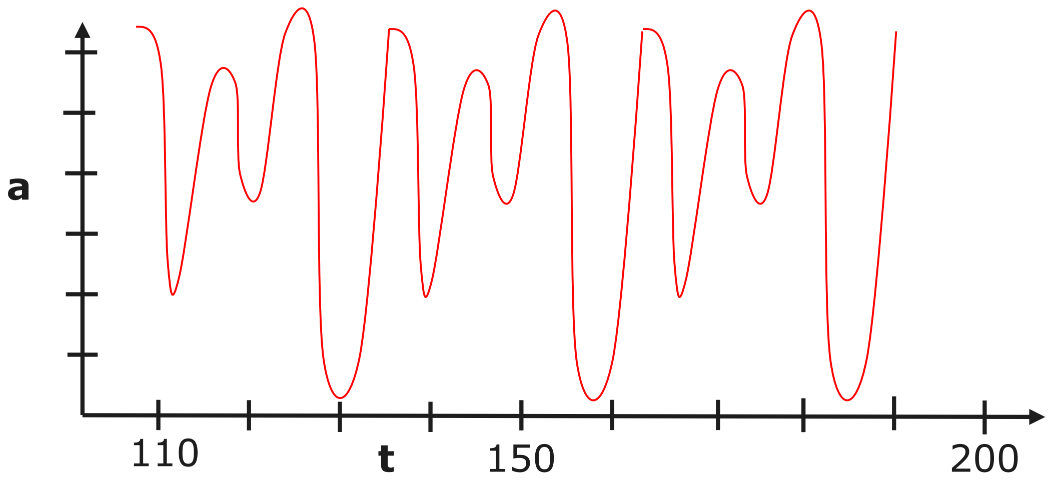
Independent variable – Always in lower case

A real valued function of a real valued independent variable

Example



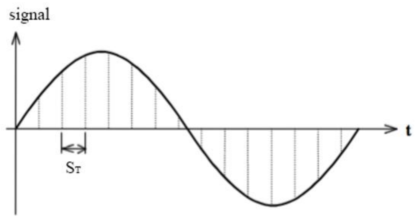
Aperiodic



Periodic

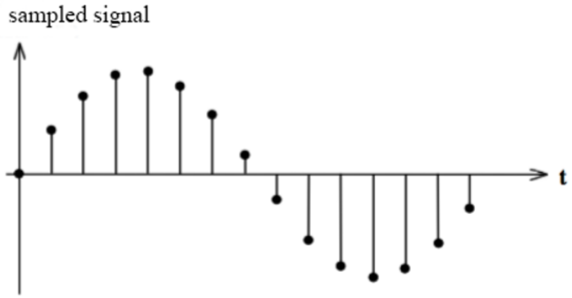
$$x(t + T) = x(t)$$
$$x(t + mT) = x(t)$$

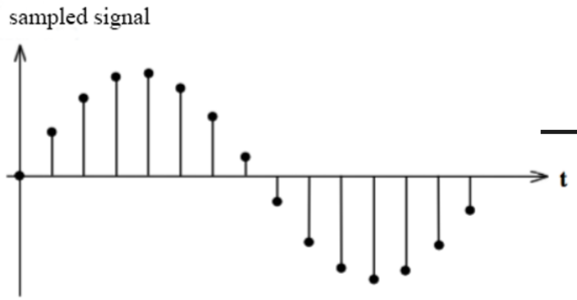
Discrete to Digital



Sample and Hold Circuit

Sampling Frequency





Quantizer and Encoder

Digital Data

0,2,3,4,5,6,7,….

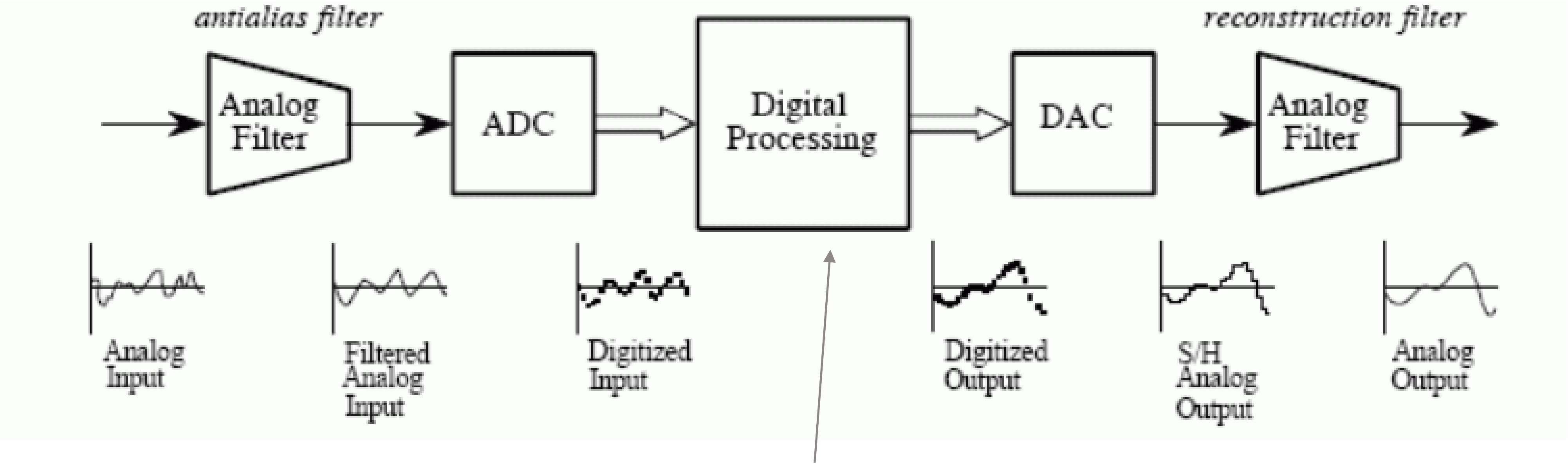
y[n]

Note Note Note Note and Note

Digital data samples DO NOT have any element of time information

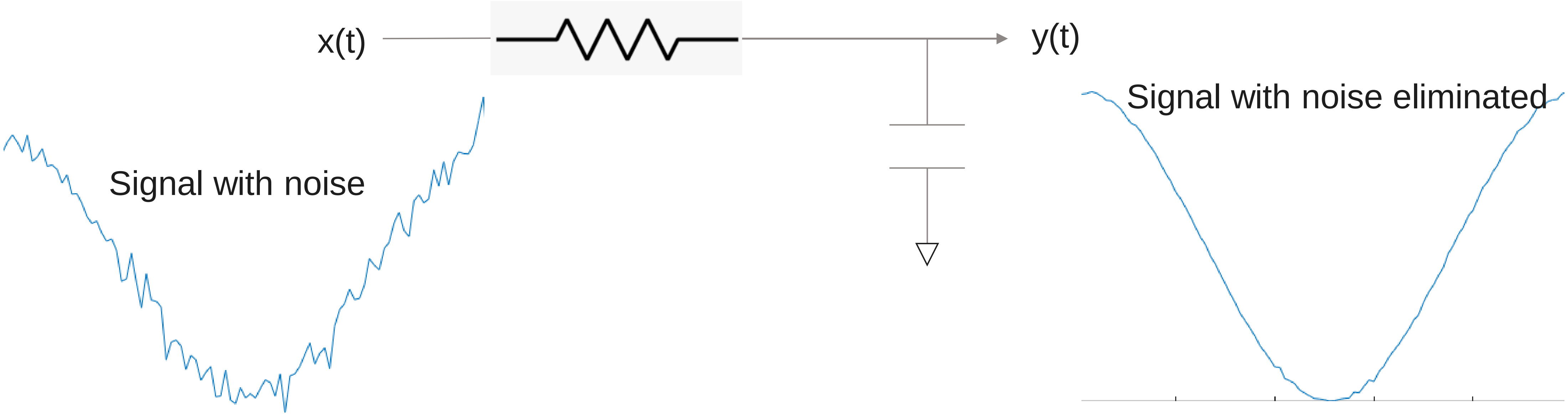
Values are no more continuous. They are discrete.

Typical signal processing chain



- Digital filtering (Windowing, FIR, IIR)
- Spectral analysis (DFT)
- Power analysis (PSD)
- Rate change (Interpolation, Decimation)
- Machine Learning (Feature extraction)

A simple analog filter



1. Voltage across a capacitor cannot change instantaneously
2. Capacitor **takes time** to charge (or discharge) to a new voltage
3. The change in capacitor voltage is gradual
4. It is because of this gradual change, the High Frequency noise is eliminated from the signal!

Voltage change across capacitor – 1/2

1. Voltage across a capacitor cannot change instantaneously

2. Capacitor takes time to charge (or discharge) to a new voltage

…

Voltage change equation $V(t) = V\_Final + (V\_Initial - V\_Final)\, e^{-t/T}$ **T is known as the Time constant = RC**

Time Constant	RC Value	Percentage of Maximum	
		Voltage	Current
0.5 time constant	0.5T = 0.5RC	39.3%	60.7%
0.7 time constant	0.7T = 0.7RC	50.3%	49.7%
1.0 time constant	1T = 1RC	63.2%	36.8%
2.0 time constants	2T = 2RC	86.5%	13.5%
3.0 time constants	3T = 3RC	95.0%	5.0%
4.0 time constants	4T = 4RC	98.2%	1.8%
5.0 time constants	5T = 5RC	99.3%	0.7%

$$t = -RC \ln\left(\frac{V(t) - V_{FINAL}}{V_{Initial} - V_{FINAL}}\right)$$

Voltage change across capacitor – 2/2

$$t = -RC \ln\left(\frac{V(t) - V_{FINAL}}{V_{Initial} - V_{FINAL}}\right)$$

If V_Initial = 3V and V_Final = 3.1V, how long does it take for the capacitor voltage to reach 99.7% of V_Final?

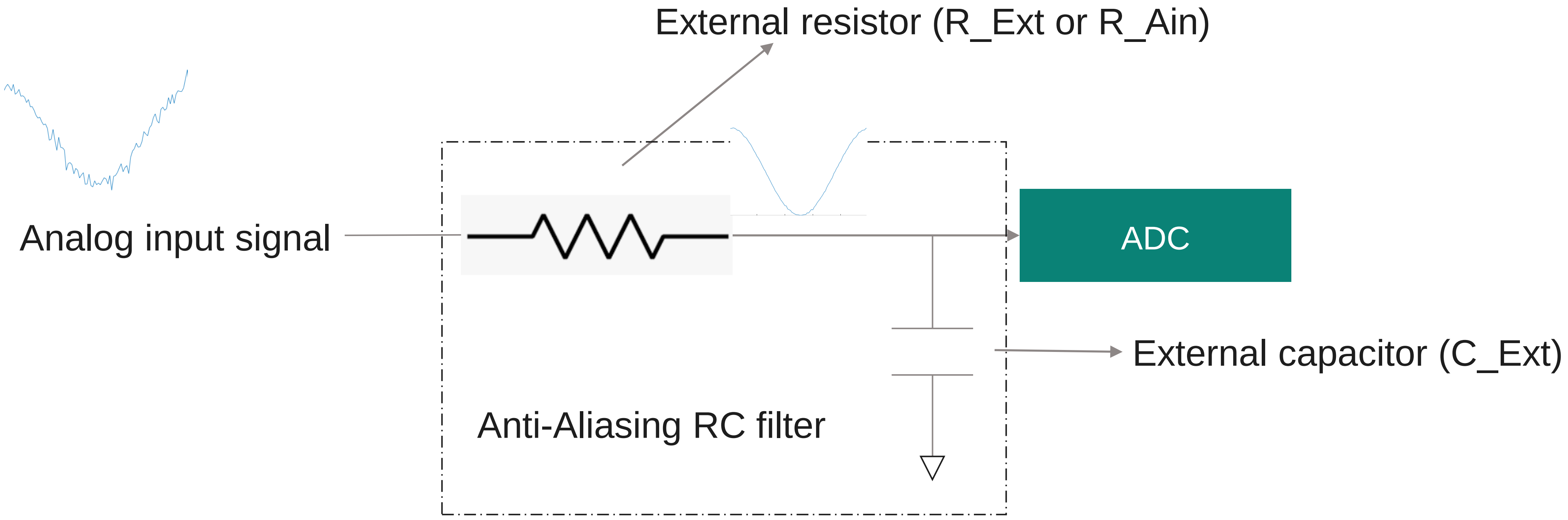
If V_Initial = 3V and V_Final = 3.5V, how long does it take for the capacitor voltage to reach 99.7% of V_Final?

If V_Initial = 3V and V_Final = 4V, how long does it take for the capacitor voltage to reach 99.7% of V_Final?

If V_Initial = 3V and V_Final = 12V, how long does it take for the capacitor voltage to reach 99.7% of V_Final?

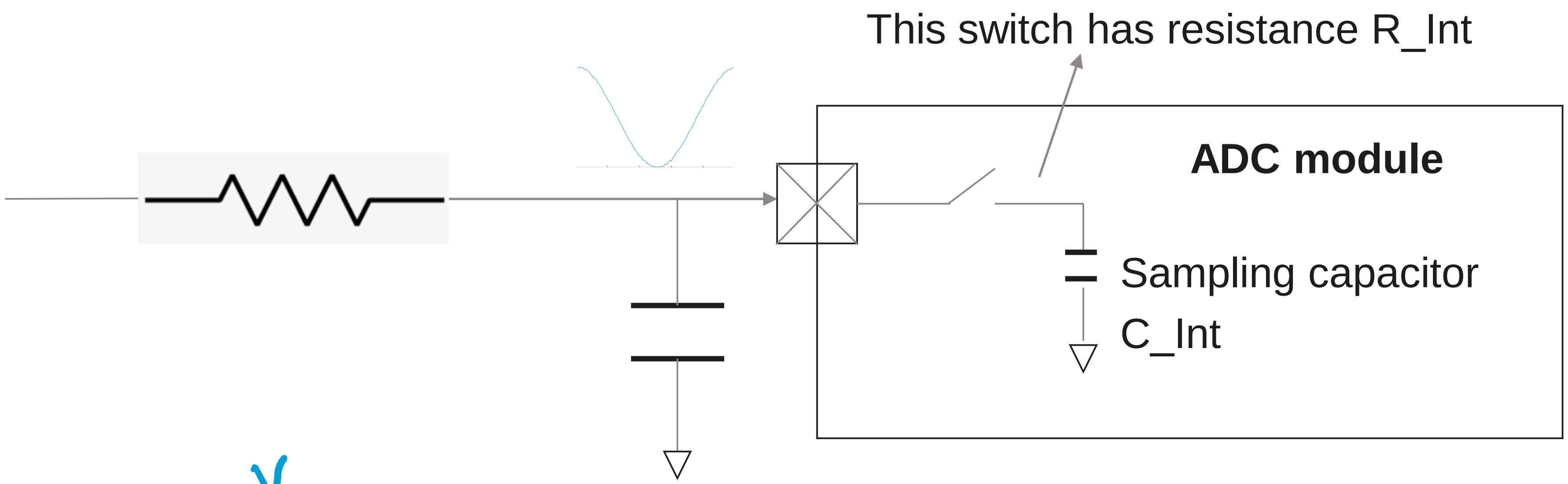
How many time constants were needed for the smallest change?

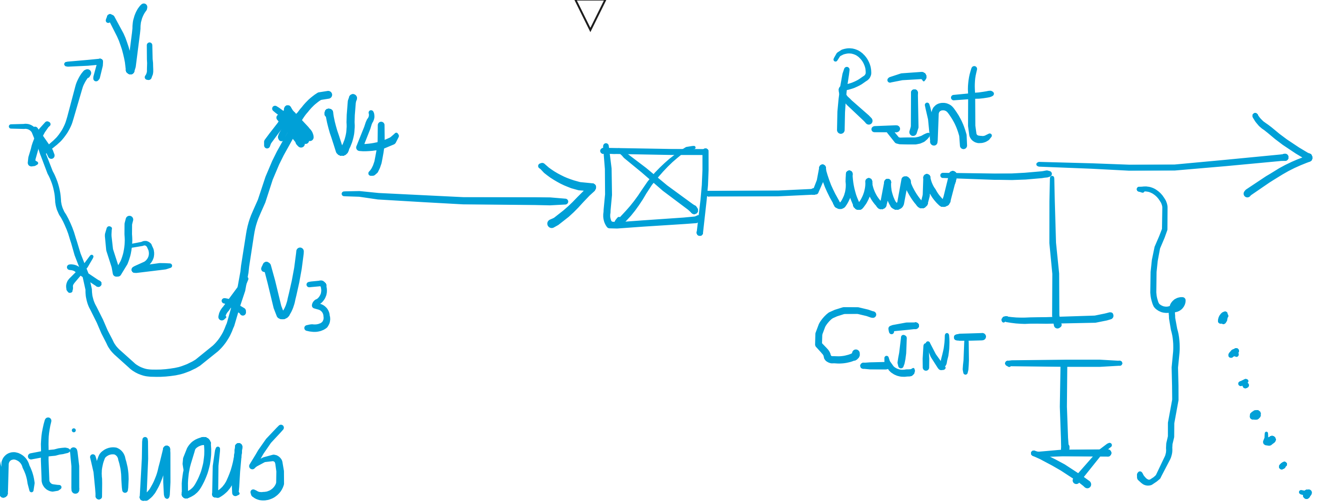
ADC input stage



Cut-off frequency Fc = 1/(2 Pi R_Ext C_Ext)

ADC Sampling stage – 1/3



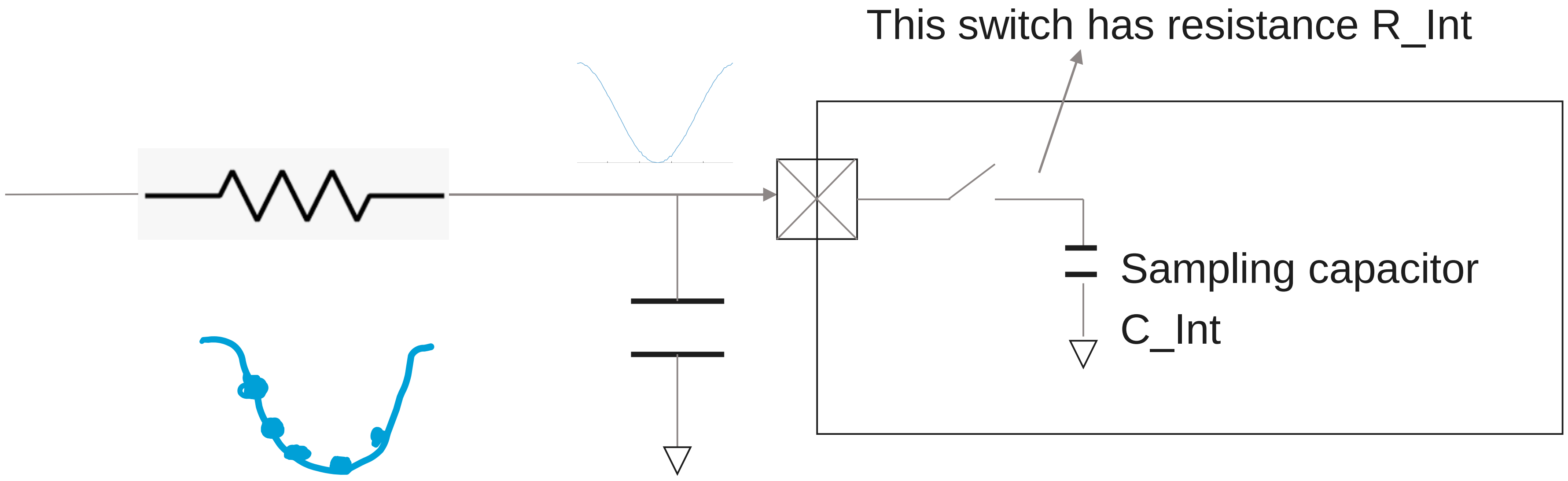


Continuous

Sampling is about charging C_Int to voltages V_1, V_2, V_3 ---

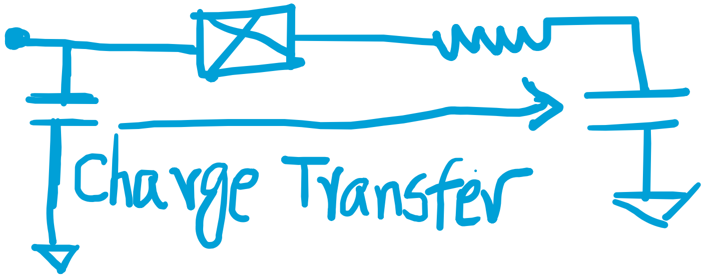
Discrete (Not Digital yet!)

ADC Sampling stage – 2/3



To acquire a sample

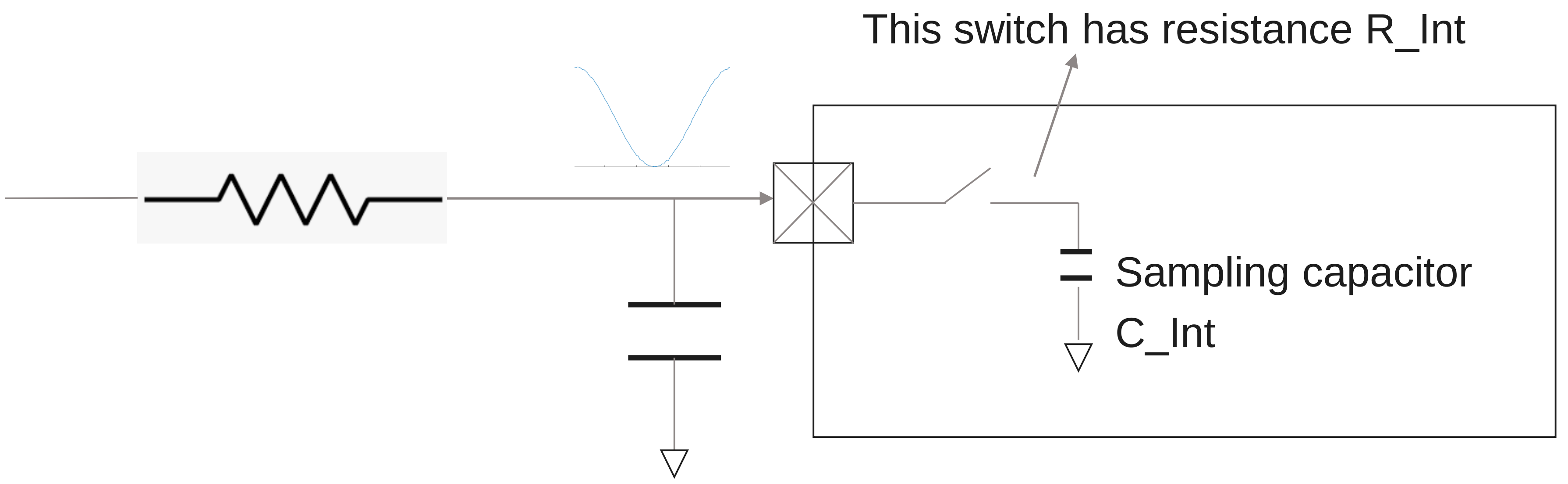
Close the switch



To Hold the Sample

Isolate C_INT by reopening the switch

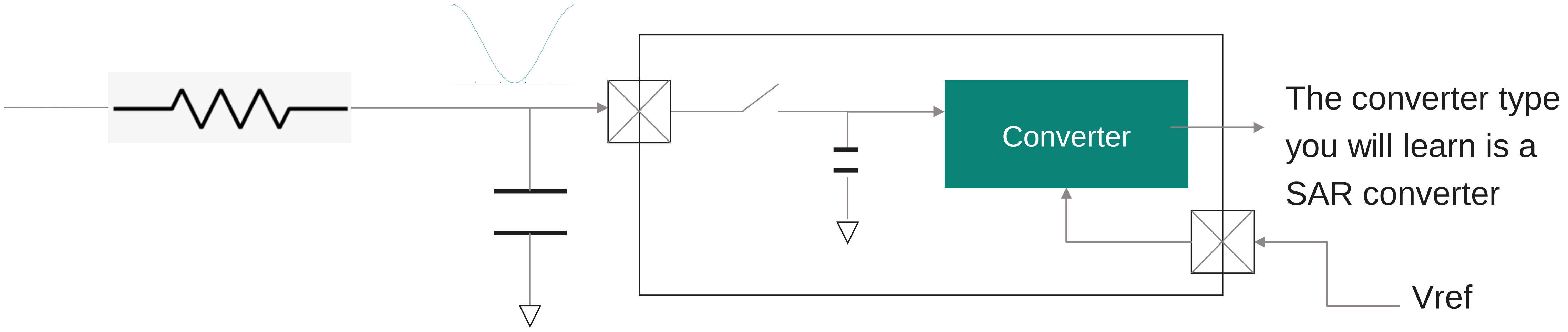
ADC Sampling stage – 3/3



R_Int and C_Int are specified by the chip manufacturer

Read the datasheet!

Conversion – 1/2



Converter reads the voltage across C_Int and converts it into a **digital code word**

Resolution of a converter determines how many code words ca be generated

Example, if the resolution is 12 bits, then 2^12 = 4096 code words (0 to 4095) can be generated by the converter

All converters need a **Reference Voltage** Vref for performing **Ratiometric** conversion

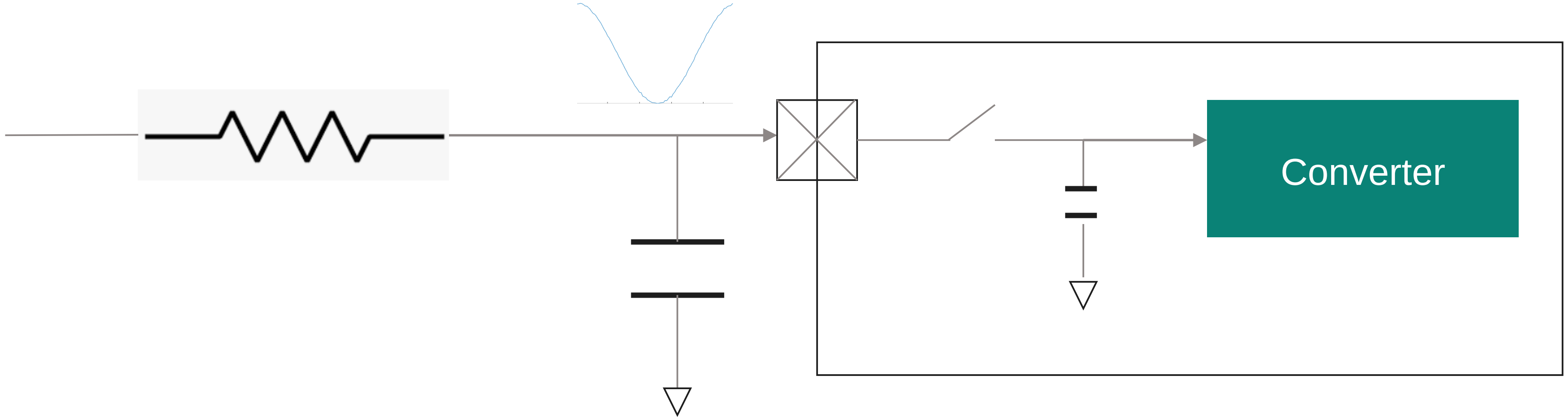
Example

Input signal is expected to be in the range 0 to 3.2V. The reference voltage supplied to ADC is 3.3 V.

A sample is taken and its value is determined as 2.85V.

Code Word = (2.85/3.3) x 4096 = 3534

Conversion – 2/2



LSB of an ADC converter is the voltage difference needed to change the code by 1

LSB = Vref/2^N where N is the resolution of the ADC converter

e.g. A 1.8V 10 bit ADC converter's LSB = 1.8/1024 = 1.76 mV

Suppose 0.9 volts produced a code of 510, then 0.9-1.76mv will lead to a code of 509

If the SAR ADC produces a code of 1187, what will have been the input voltage?

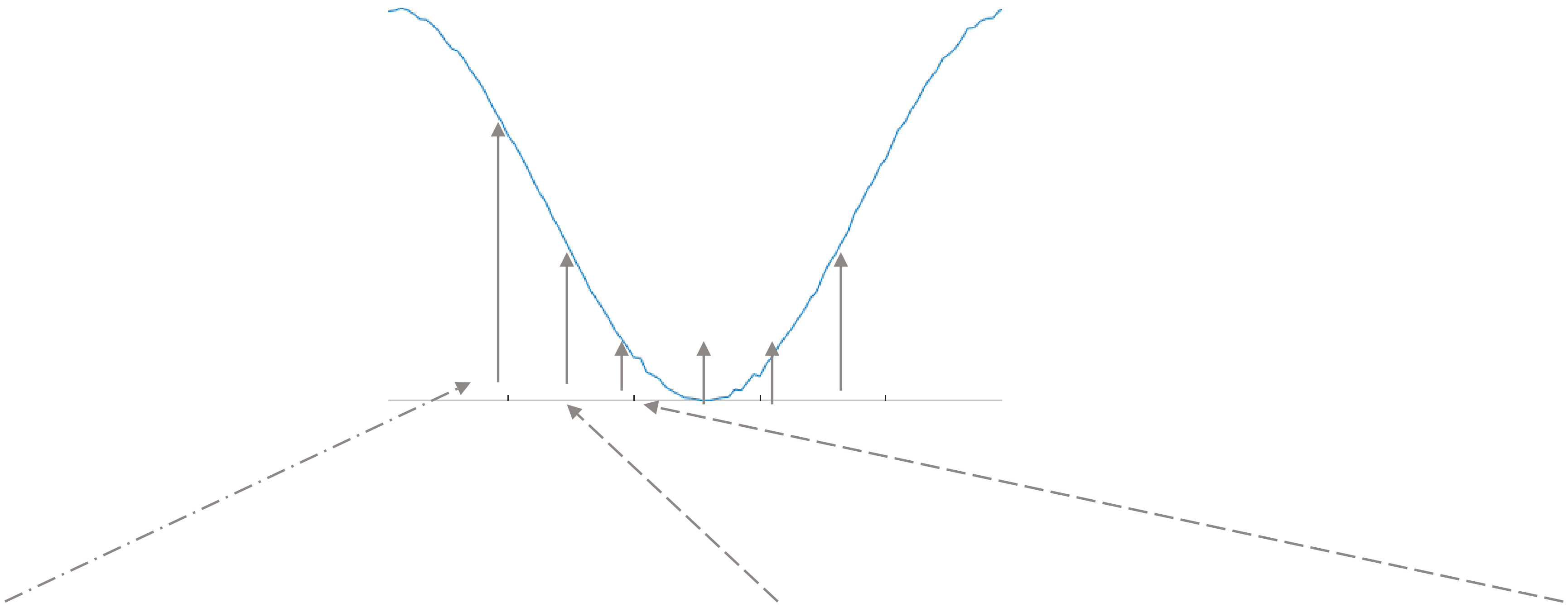
A 12 bit converter takes 12 clock cycles to generate a 12 bit code word

A 10 bit converter takes 10 clock cycles to generate a 10 bit code word

Homework

- Understand and explain in your own words the following additional properties of a SAR converter
  - Offset error
  - DNL
  - INL
  - TUE
  - THD
  - SINAD
  - ENOB

Procedure to determine ADC clock frequency



Sample and Hold Time	Conversion time		Sample and Hold Time	Conversion time		Sample and Hold Time	….

10 RC

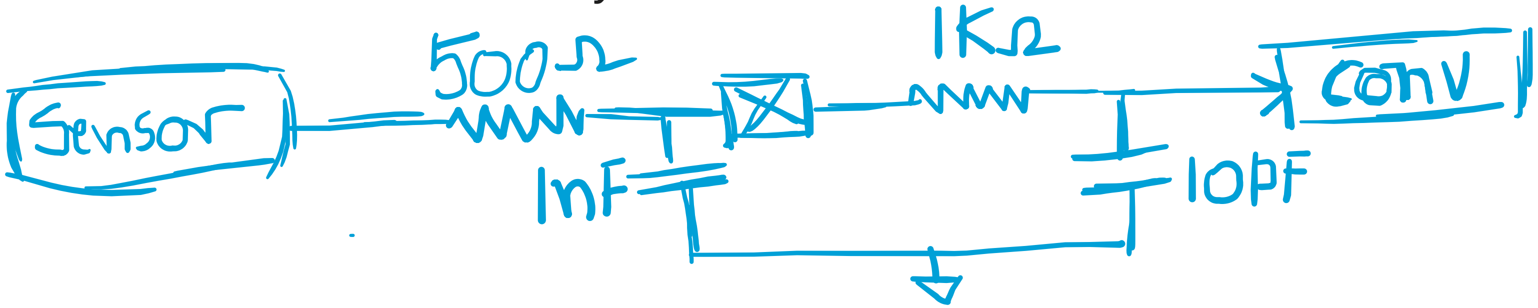
12 clock cycles

Procedure to determine ADC clock frequency

Sample and Hold Time	Conversion time		Sample and Hold Time	Conversion time		Sample and Hold Time	….

10 RC

12 clock cycles



Cut off frequency = 1/(2 x Pi x 500 x 1nF) = 320 KHz

Sampling frequency > 640 KHz ➔ 1000 KHz (1 MHz)

Sampling interval = 1/1 MHz = 1 uS

Time constant of the S&H circuit = 1K x 10pF = 10^{-8} = 100 ns

Procedure to determine ADC clock frequency

Highest input signal frequency = 80 KHz

Sampling frequency > 160 KHz rounded to 500 KHz, Ts = 2 uS

Sample and Hold Time	Conversion time		Sample and Hold Time	Conversion time		Sample and Hold Time	….

10 RC   12 clock cycles

Start with an assumption that 10 RC (10 x 10^{-8} S) requires 10 clock pulses of an unknown ADC clock

12 clock pulses for conversion is fixed

22 clock pulses in 2 uS leads to ADC frequency of 11 MHz

We need an integer divided clock.

Let us revise our estimate of S&H quota to 4.

So that will mean a total of 16 clock pulses in 2 us sampling window. That gives us 8 MHz (Good)

4 clock pulses for S&H at 8 MHz amounts to 500 ns

10 Time constants of S&H circuit = 10 x 10^{-8} = 100 ns

Real S&H time of 500 ns exceeds the minimum 100 ns S&H requirement. We stick to 8 MHz as ADC frequency!

Next class: PSOC ADC programming model

Infineon